AN EXACT CERTIFICATE ABOVE 2.032083 FOR SENSITIVITY VERSUS BLOCK SENSITIVITY

Abstract. An exact integer certificate for an existing nested-tournament construction gives an explicit infinite family of total Boolean functions with unbounded ordinary sensitivity and block sensitivity at least sensitivity to a fixed power $\alpha > 2.032083$. The proof includes the finite deterministic label search, the indexed all-input sensitivity bounds, disjoint block witnesses, and composition at every depth. It uses OpenAI's construction and ProveIt Part I's twenty-stage schedule and profile analysis, retaining the exact final sensitivity envelope and taking 99 OR copies. A dependency-free integer checker establishes $\alpha > 380/187 > 2.032083$; it does not replace the combinatorial proof.

1. Statement, attribution, and scope

For $x \in \{0,1\}^n$ and a coordinate set D, let x^D flip precisely the coordinates in D, and write $x^a = x^{\{a\}}$. For a total Boolean function,

$$s(f,x) = |\{a : f(x^a) \neq f(x)\}|, \qquad s(f) = \max_x s(f,x).$$

Let $s_b(f) = \max_{x:f(x)=b} s(f,x)$, with an empty maximum equal to zero. The block sensitivity $\mathrm{bs}(f,x)$ is the maximum number of pairwise disjoint nonempty coordinate sets D whose separate flips change $f(x)$; put $\mathrm{bs}(f) = \max_x \mathrm{bs}(f,x)$. All sensitivities below are these ordinary worst-case measures, not spectral or average sensitivities.

The nested-predicate architecture is due to the September 25, 2026 OpenAI manuscript [1]. The level-dependent schedule and indexed profile inequalities used here are from ProveIt Part I, dated October 7, 2026 [2]. Their proofs are included rather than assumed. The certificate retains Part I's exact sensitivity envelope and uses 99 OR copies instead of 100. This gives an exact certificate for the construction.

Fix

$$\begin{aligned} d = 20, \qquad L = 22, \qquad b_0 = 2^{18}, \qquad m_0 = 99, \\ (e_1,\dots,e_{20}) = (43,43,39,39,40,40,37,36,34,32, \\ 30,28,26,24,23,22,20,19,18,15), \\ h_\ell = 2^{e_\ell}, \qquad k_\ell = 2h_\ell + 1, \qquad \tau_\ell = e_\ell + 6. \end{aligned} \tag{1}$$

Define the following positive integers, writing S as two decimal pieces to make its full value legible:

$$\begin{aligned} B &= 99 \cdot 2^{18} \prod_{\ell=1}^{20} (2^{e_\ell+1}+1), \\ S &= 4719112961253199416318700714565811194436816937 06 \cdot 10^{49} \\ &\quad + 90424261434806749864886539252667781359055446999 04, \\ n &= 99 \cdot 22 \cdot 2^{18} \cdot 4^{20} \prod_{\ell=1}^{20} k_\ell, \qquad \alpha = \frac{\log B}{\log S}. \end{aligned} \tag{2}$$

Date: October 17, 2026.

Theorem 1.1. *There is an explicitly specified total Boolean function f on n bits such that, for $F_1 = f$ and $F_{m+1} = f \circ F_m$ on disjoint inputs, every integer $m \geq 1$ satisfies*

$$F_m : \{0,1\}^{n^m} \longrightarrow \{0,1\}, \qquad F_m(\mathbf{0}) = 0,$$
$$99^m \leq \min\{s_0(F_m), s_1(F_m)\} \leq s(F_m) \leq S^m, \tag{3}$$
$$\mathrm{bs}(F_m, \mathbf{0}) \geq B^m \geq s(F_m)^\alpha, \qquad \alpha > \frac{380}{187} > 2.032083.$$

Thus sensitivity tends to infinity and $\mathrm{bs}(F_m) \geq c\, s(F_m)^\alpha$ with the explicit constant $c = 1$.

The specification is computable by terminating finite searches.

2. A finite deterministic construction

2.1. **Tournament labels.** For each scheduled integer e, put $h = 2^e$, $k = 2h + 1$, and $\tau = e + 6$. On vertices $0, \ldots, k-1$, direct $i \to j$ when $(j - i) \bmod k$ belongs to $\{1, \ldots, h\}$. This is a regular tournament of outdegree h. Give each directed edge a label $a(i,j) \in \{1,2,3,4\}$. Require that there be no τ-element set I and choices $m_j \in \{1,2,3,4\}$, $j \in I$, for which

$$a(i,j) = m_j \quad \text{for every edge } i \to j \text{ within } I. \tag{4}$$

Such labels exist: independent uniform labels satisfy the conditions for fixed $I, (m_j)$ with probability $4^{-\binom{\tau}{2}}$. The union bound over at most $k^\tau 4^\tau$ choices is

$$k^\tau 4^{\tau - \binom{\tau}{2}} < 2^{\tau(e+2) + 2\tau - \tau(\tau-1)} = 2^{\tau(e+5-\tau)} = 2^{-\tau} < 1.$$

If $\tau > k$ the requirement is vacuous. Order the directed edges lexicographically by (i,j) and choose the first lexicographic label vector passing the finite exclusion test. Existence proves that this deterministic search terminates. Use this rule separately at every level.

2.2. **Nested predicates and the recursive interface.** Write $H_\ell = d + 1 - \ell$. At level ℓ there are H_ℓ predicates on

$$N_\ell = L b_0\, 4^\ell \prod_{j=1}^{\ell} k_j$$

coordinates, nested as $P_{\ell,1} \geq \cdots \geq P_{\ell,H_\ell}$. At level zero, partition the coordinates into b_0 consecutive blocks E_j of size L. Let z_j be their indicator vectors and set

$$D(x) = \min_j \mathrm{dist}(x, z_j), \qquad P_{0,q}(x) = \mathbf{1}\{D(x) \leq H_0 - q\}.$$

Here dist is Hamming distance. Distinct centres are at distance $2L$. The radii range from d to zero, so nesting holds, all predicates reject at zero since $D(\mathbf{0}) = L > d$, and all accept at each z_j.

At level $\ell \geq 1$, split the input into disjoint children $x_{j,c}$, one $N_{\ell-1}$-bit string for each row $j \in \{0, \ldots, k_\ell - 1\}$ and position $c \in \{1,2,3,4\}$. Put $Q = H_{\ell-1} = H_\ell + 1$. For each $1 \leq q \leq H_\ell$, clause i requires

$$P_{\ell-1,Q}(x_{i,c}) = 1 \qquad (c = 1,2,3,4), \tag{5}$$
$$P_{\ell-1,Q-q}(x_{j,a(i,j)}) = 0 \qquad (i \to j). \tag{6}$$

Define $P_{\ell,q}$ as the OR of all k_ℓ clauses. Conditions (5) are targets and (6) are gates. Each condition in a fixed clause concerns a different child. Thus one original bit flip changes at most one of its conditions, although it can affect several clauses.

The child gate index obeys $1 \leq Q - q < Q$, so every called predicate exists and differs from the strongest target predicate. Increasing q decreases $Q - q$, increasing the child gate predicate by nesting and making its zero condition more restrictive. This proves nesting inductively. The definitions give total Boolean functions with the stated dimensions and a valid child interface at every level.

3. Sensitivity bounds at all inputs

Let $S_{\ell,b}(q) = s_b(P_{\ell,q})$. Let $J_{\ell,b}$ be the maximum, over $p < q$ and inputs with $P_{\ell,p}(x) = P_{\ell,q}(x) = b$, of the number of coordinates whose individual flip changes both predicates to $1 - b$. An empty maximum is zero. The same coordinate and input are used for both predicates.

Lemma 3.1. *At level zero,*

$$S_{0,0}(q) \le L - q, \qquad S_{0,1}(q) \le Lb_0 - d - 1 + q, \qquad J_{0,0} = J_{0,1} = 0. \tag{7}$$

Proof. Put $R = d + 1 - q$. At a rejecting input, an accepting flip approaches a centre at distance $R + 1 \le d + 1$. Two such centres would be at distance at most $2d + 2 < 2L$, so at most one exists and at most $R + 1 = L - q$ flips approach it. At an accepting input inside an R-ball, no flip leaves the ball unless the input is on its boundary. At the boundary at most $Lb_0 - R$ flips increase the distance; other accepting balls can only decrease the number of rejections. Finally D is 1-Lipschitz on cube edges. Two distinct integral radius thresholds cannot both be crossed in the same direction by one bit flip. This gives the two joint bounds. □

For one recursive step, suppress the child level index, use primes for parent profiles, and put $h = h_\ell$, $\tau = \tau_\ell$, $Q = H_{\ell-1}$, and $g = Q - q$.

Proposition 3.2 (Indexed profile inequalities, ProveIt Part I)**.** *For every valid parent index q,*

$$S_1'(q) \le hS_0(g) + 4S_1(Q), \tag{8}$$

$$S_0'(q) \le (\tau - 1)S_0(Q) + \max\{S_1(g) + J_1, 3J_1\}, \tag{9}$$

$$J_1' \le hJ_0 + 4S_1(Q), \tag{10}$$

$$J_0' \le (\tau - 1)S_0(Q) + 3J_1. \tag{11}$$

Proof. At an accepting input, fix one accepting clause. A flip making the parent reject must destroy a target or a gate of that clause. Its four targets contribute at most $4S_1(Q)$ and its h gates at most $hS_0(g)$. This proves (8).

At a rejecting input, form two fixed lists: T consists of clauses with exactly one failed target and no failed gate; G consists of clauses with exactly one failed gate and no failed target. Any clause repaired by one bit belongs to one of these lists because its conditions use different children.

For $j \in T$, let m_j denote its unique failed target position. If $i, j \in T$ and $i \to j$, clause i's gate in row j has value zero. Nesting forces the strongest child predicate there to be zero as well. It is therefore row j's unique failed target, so $a(i, j) = m_j$. Condition (4) implies $|T| \le \tau - 1$. These clauses contribute at most $(\tau - 1)S_0(Q)$ possible flips.

Every row in G has all four targets true. Hence every edge within G is a failed gate of its tail clause. Each induced outdegree is at most one. With $u = |G|$, counting tournament edges gives $u(u - 1)/2 \le u$, so $u \le 3$. If $u = 3$, each outdegree is one; if $u = 2$, there is a source and a sink.

Whenever a row in G has an outgoing edge within G, this is its only failed gate. The selected child initially has both predicates g and Q equal to one. Repairing the gate makes $g = 0$, and nesting forces $Q = 0$. Since $g < Q$, there are at most J_1 such flips. Thus $u = 3$ contributes at most $3J_1$; for $u = 2$ the source contributes J_1 and the sink at most $S_1(g)$; for $u = 1$ the bound is $S_1(g)$; for $u = 0$ it is zero. Adding the T contribution proves (9). These union bounds do not presume simultaneous attainability of maxima.

For (10), fix $q < q'$ at an input where both parents accept. A clause accepting for q' also accepts for q. Write $g' = Q - q' < g = Q - q < Q$. A flip rejecting both parents must break a target of this clause, with at most $4S_1(Q)$ choices, or break a gate even for q. At this gate initially $g' = g = 0$, and afterward $g = g' = 1$, by nesting. Each gate therefore contributes at most J_0, giving the claimed bound.

For (11), fix $q < q'$ at an input where both parents reject. Use T, G for the weaker parent q. For each successful flip, choose a clause accepting afterward for q'; it also accepts for q and so was in T

or G. Clauses in T contribute at most the target bound already proved. A clause in G initially has $g = 1$, hence $g' = 1$, at its failed gate. To accept for q' afterward it must have $g' = 0$, hence $g = 0$. Each of the at most three clauses contributes at most J_1, including a possible sink. Additional failed conditions for q' only remove successful flips. Taking maxima proves all four inequalities. If there is only one parent predicate, the joint maxima are empty and their bounds still hold. □

Define nonnegative integer envelopes by

$$A_{0,q} = L - q, \quad C_{0,q} = Lb_0 - d - 1 + q, \quad U_0 = V_0 = 0.$$

At level $\ell \geq 1$, put $Q = d + 2 - \ell$ and, for $1 \leq q < Q$, define

$$\begin{aligned} A_{\ell,q} &= (\tau_\ell - 1)A_{\ell-1,Q} + \max\{C_{\ell-1,Q-q} + V_{\ell-1}, 3V_{\ell-1}\}, \\ C_{\ell,q} &= h_\ell A_{\ell-1,Q-q} + 4C_{\ell-1,Q}, \\ U_\ell &= (\tau_\ell - 1)A_{\ell-1,Q} + 3V_{\ell-1}, \\ V_\ell &= h_\ell U_{\ell-1} + 4C_{\ell-1,Q}. \end{aligned} \tag{12}$$

Lemma 3.1, Proposition 3.2, and monotonicity of the right-hand sides in all their arguments give by induction

$$S_{\ell,0}(q) \leq A_{\ell,q}, \quad S_{\ell,1}(q) \leq C_{\ell,q}, \quad J_{\ell,0} \leq U_\ell, \quad J_{\ell,1} \leq V_\ell. \tag{13}$$

These hold at every input, not only near the block witnesses below. Targets use the last child index Q, and gates use $Q - q$ throughout. Positive final joint envelopes cause no problem even though the actual final joint maxima are zero.

4. Block witnesses and the exact seed certificate

Lemma 4.1. *At level ℓ, all predicates vanish at zero. There are $b_\ell = b_0 \prod_{j=1}^{\ell} k_j$ disjoint nonempty coordinate blocks, each of size $L4^\ell$, whose individual flips at zero make every level-ℓ predicate equal to one.*

Proof. The base blocks are E_j. Inductively, for each row and each previous block, take the union of its copies in the four children of that row. Flipping it makes that row's targets true, while its gates lie in other rows and remain zero. Its clause accepts for every predicate index. Different previous blocks remain disjoint within each child, and different rows use disjoint children. This proves the number and size of blocks. At zero all targets fail, so every predicate rejects. □

Let f be the OR of 99 disjoint copies of $P_{20,1}$. Its dimension is n from (2), it vanishes at zero, and the blocks in Lemma 4.1 show that it is nonconstant and that

$$\mathrm{bs}(f, \mathbf{0}) \geq B, \qquad s(f) \leq \max\{99A_{20,1}, C_{20,1}\}. \tag{14}$$

Indeed, at a rejecting OR input each copy contributes at most $A_{20,1}$. At an accepting input with at least two accepting copies, no single-bit flip rejects the OR. With exactly one accepting copy, all sensitive coordinates lie there and contribute at most $C_{20,1}$.

The supplied `checkers/verify_certificate.py` evaluates precisely (12) with arbitrary-precision integers and verifies

$$\begin{gathered} C_{20,1} = S, \qquad 99A_{20,1} \leq S, \qquad B^{187} > S^{380}, \\ 380 \cdot 10^6 > 2032083 \cdot 187. \end{gathered} \tag{15}$$

It also checks the union-bound label inequality at all twenty stages and the final block count. Its checks remain enabled under Python's optimization flag. This finite calculation gives $s(f) \leq S$ and $\alpha > 380/187 > 2.032083$ exactly; no floating-point logarithm is used. The checker neither constructs the huge label tables nor establishes the all-input inequalities; those are the preceding proofs.

5. Sensitivity growth and composition at every depth

5.1. **Both output-conditioned sensitivities grow.** At level zero the strongest predicate has radius zero. Every one of the Lb_0 coordinates at z_1 is sensitive for it, since no other centre is at distance one from z_1. Put $W_0 = z_1$. Define W_ℓ by putting $W_{\ell-1}$ into all four children of row zero and zero into all other rows. Every predicate accepts at W_ℓ: row zero's targets hold and its gates remain zero.

Inductively the strongest predicate has at least $4^\ell Lb_0$ sensitive coordinates at W_ℓ. A designated sensitive flip in any one of its four target children breaks row zero's clause. Every other row retains four false targets because its children stay zero. Thus the strongest parent rejects. The four child coordinate sets are disjoint, proving the induction.

In particular $P_{20,1}$ accepts at W_{20} with at least $R = 4^{20}Lb_0$ sensitive coordinates. Flip one such coordinate to get a rejecting input Y with a sensitive return flip. At 99 copies of Y, the seed OR rejects and has a sensitive coordinate in each copy. At one copy of W_{20} and 98 zero copies, the seed accepts and all R designated coordinates make it reject. Consequently

$$\delta(f) := \min\{s_0(f), s_1(f)\} \geq 99. \tag{16}$$

These witnesses are also deterministic, for example by choosing the least designated coordinate when defining Y.

5.2. **Composition identities and completion of the proof.** For total f on a bits and g on b bits, write $(f \circ g)(x_1, \ldots, x_a) = f(g(x_1), \ldots, g(x_a))$, with disjoint children, and put $y_i = g(x_i)$. A bit in child i changes the output exactly when it changes $g(x_i)$ and i is sensitive for f at y. Hence

$$s(f \circ g, x) = \sum_{i : f(y^i) \neq f(y)} s(g, x_i) \leq s(f)s(g). \tag{17}$$

If $g(\mathbf{0}) = 0$, take disjoint sensitive blocks $C_1, \ldots, C_p$ for f at zero and $D_1, \ldots, D_q$ for g at zero. In product coordinates the pq sets $C_i \times D_j$ are nonempty and disjoint. Flipping one makes exactly the children in C_i output one and so changes the outer value. Therefore

$$\mathrm{bs}(f \circ g, \mathbf{0}) \geq \mathrm{bs}(f, \mathbf{0})\,\mathrm{bs}(g, \mathbf{0}), \qquad (f \circ g)(\mathbf{0}) = f(\mathbf{0}). \tag{18}$$

For nonconstant f, g, choose y attaining $s_b(f)$ with $f(y) = b$. For each i, choose x_i attaining $s_{y_i}(g)$ with $g(x_i) = y_i$. Both outputs of g exist. The exact sum in (17) gives

$$s_b(f \circ g) \geq s_b(f)\delta(g), \qquad \delta(f \circ g) \geq \delta(f)\delta(g). \tag{19}$$

For $F_1 = f$, $F_{m+1} = f \circ F_m$, induction using (14)–(19) now gives

$$F_m(\mathbf{0}) = 0, \quad \delta(F_m) \geq 99^m, \quad s(F_m) \leq S^m, \quad \mathrm{bs}(F_m, \mathbf{0}) \geq B^m.$$

The zero value and the block witnesses show nonconstancy at every step, so the lower composition bound always applies. Since $S^\alpha = B$,

$$\mathrm{bs}(F_m) \geq \mathrm{bs}(F_m, \mathbf{0}) \geq B^m = (S^m)^\alpha \geq s(F_m)^\alpha.$$

Also $s(F_m) \geq 99^m \to \infty$. Together with (15), this proves Theorem 1.1.

6. Improved exponent

OpenAI's displayed seed gives exponent approximately 2.00025240714; its published estimates at depth 20 give approximately 2.00647082823 [1, 2]. The construction and exact sensitivity certificate above yield

$$\alpha > 380/187 > 2.032083.$$

The indexed schedule and profile inequalities follow ProveIt Part I. Retaining its exact sensitivity envelope and choosing 99 OR copies produces the finite certificate used here. Composition extends it to an infinite family with unbounded sensitivity.

References

1. OpenAI, *A superquadratic separation between sensitivity and block sensitivity*, OpenAI Math Release preprint, September 25, September 2026, Revision 1581d4cbdd69a0faf34b2089614fe2a814316a6a; bibliographic record supplied by the repository README; pinned primary source.
2. *An explicit exponent above* 65/32 *for ordinary sensitivity versus block sensitivity*, ProveIt repository, Part I, October 7, October 2026, Repository companion; revision 5476940edc3a2e7ea0c5426f013c6b5966451d73; pinned primary source.

Mathematical supplement

The exact arithmetic programs are included in the attached mathematical supplement (ZIP). Open the attachment in a PDF reader with file-attachment support.